

Rahul Nomula

Data Engineer

Denton, TX | rahulnomula03@gmail.com | +1 (408) 390 6791

PROFESSIONAL SUMMARY

Over 4 years of experience as a Data Engineer, specializing in advanced technologies for data-driven decision-making. Proficient in Python, SQL, Scala, and big data tools like Apache Spark, Hadoop, and Kafka. Experienced in building and optimizing ETL pipelines with Apache Airflow, Kafka, and Spark. Skilled in creating data visualizations with Tableau and Power BI, turning complex datasets into actionable insights. Expertise in cloud platforms (AWS, Azure, GCP) and database optimization (SQL: PostgreSQL, NoSQL: MongoDB, Cassandra). Proficient in applying machine learning and statistical analysis to extract insights from large datasets.

EXPERIENCE

Humana, USA | Data Engineer

Jan 2024 - Present

- Spearheaded Agile, cross-functional teams in developing and deploying data pipelines, resulting in a 40% increase in project delivery speed and a 30% improvement in code quality.
- Engineered and optimized data processing frameworks using Hadoop, MapReduce, and Apache Spark, achieving a 50% reduction in processing time for large-scale datasets exceeding 10TB.
- Architected and optimized ETL workflows using Python, SQL, SSIS, and Talend, enhancing data throughput and system efficiency.
- Conducted advanced data cleansing and wrangling with Pandas, PySpark, and SQL, ensuring high data quality and consistency for machine learning and reporting purposes.
- Applied machine learning tools (such as Azure Machine Learning, TensorFlow, and Scikit-learn) for analyzing large health insurance datasets, improving claim prediction accuracy and patient outcome forecasting by 30%.
- Developed and maintained real-time monitoring dashboards using Tableau, providing 99.9% uptime visibility for 15+ machine learning models and 20+ data pipelines.
- Migrated data from on-premises to Azure Blob Storage and Synapse Analytics, ensuring minimal downtime and data integrity.
- Collaborated with teams to deploy end-to-end data solutions using Azure Databricks and Azure DevOps, enabling agile project delivery.
- Engineered and maintained data warehouses in MySQL and PostgreSQL, improving data storage and retrieval performance.

Mphasis, India | Data Engineer

Aug 2018 - Jul 2022

- Led end-to-end data integration projects using the Waterfall methodology, ensuring thorough planning, seamless execution, and timely delivery of complex data pipelines.
- Designed and implemented ETL processes using Apache NiFi and Apache Airflow to automate data workflows and enhance operational efficiency.
- Utilized Pig, Sqoop, and PySpark for processing datasets over 10 TB, reducing data processing times by 35% and enabling quicker business insights.
- Architected and optimized data models and pipelines in Snowflake, improving data warehousing and retrieval processes.
- Developed data visualizations with Seaborn, ggplot2, and Power BI, leading to a 50% increase in actionable insights for business decision-making.
- Employed TensorFlow to integrate machine learning models, enabling predictive analytics and data-driven strategies.
- Designed and implemented scalable data storage and processing solutions on AWS and GCP, cutting infrastructure costs by 25% while handling a 3x increase in data volume.
- Maintained and optimized MongoDB and SQL Server databases, ensuring efficient data retrieval and storage solutions to support business operations.

SKILLS

Methodologies: SDLC, Agile, Waterfall

Programming Languages: Python, Java, Scala, R, SQL, SAS

IDE's: PyCharm, Jupyter Notebook, Visual Studio Code

Big Data Technologies: Hadoop, MapReduce, Hive, Apache Spark, Apache Hudi, Kafka, Flink, Sqoop, Pig, PySpark, Spark Structured Streaming, Delta Lake, Delta Live Tables (DLT), CDC (Change Data Capture)

ETL Tools: SSIS, Apache NiFi, Apache Kafka, Talend, Apache Airflow, Informatica, AWS Glue, Matillion, Azure Data Factory, Fivetran, DBT, Dagster

Cloud Platforms: AWS (Lambda, S3, Kinesis, Athena, Redshift, EMR, EC2), Azure (Blob Storage, Synapse Analytics, Databricks, Keyvault, DevOps), GCP (Dataproc, Cloud Storage, IAM)

Data Science Packages: NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, Seaborn, PyTorch, TensorFlow, ggplot2

Business Intelligence Tools: Tableau, Power BI, QuickSight, Microsoft Excel

Databases: SQL (MySQL, PostgreSQL, Redshift, BigQuery, Snowflake, RDS), NoSQL (MongoDB, Cassandra, HBase, DynamoDB)

Version Control: Git, GitHub, GitLab

DevOps & Containerization: Jenkins, Kubernetes, Docker, Terraform

Miscellaneous: Kimball's Dimensional Modeling, Data Cleaning, Data Wrangling, Statistical Analysis, ELK Stack, Grafana, Jira

Operating Systems: Windows, Linux

Other Skills: Critical Thinking, Communication Skills, Presentation Skills, Problem-Solving

EDUCATION

Master of Science in Data Science | University of North Texas, Denton, TX